Chemistry

1A 1	2A 2	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8	8B 9	8B 10	1B 11	2B 12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01																	2 He 4.00
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 La 138.9	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (281)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (289)	116 Lv (293)	117 Ts (293)	118 Og (294)

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Constants

R = 0.08206 L·atm/K·mol
= 8.314 J/K·mol
= 62.36 L·torr/K·mol

e = 1.602×10^{-19} C

N_a = 6.022×10^{23} mol^{-1}

k = 1.38×10^{-23} J/K

h = 6.626×10^{-34} J·s

c = 3.00×10^{8} m/s

$\mathcal{R}$ = 2.178×10^{-18} J

m_e = 9.11×10^{-31} kg

$\mathcal{F}$ = 96485 C/mol

Vapor Pressure (torr) of water

15° C	12.79
20° C	17.54
25° C	23.77
30° C	31.84

Selected Reduction Potentials

$Cu^{2+} + 2e^- \rightarrow$	Cu	0.34 V
$Zn^{2+} + 2e^- \rightarrow$	Zn	-0.76 V
$Br_2 + 2e^- \rightarrow$	$2Br^-$	1.07 V

Conversions

1 atm	= 760 torr
	= 101325 Pa
	= 14.7 psi
1 bar	= 105 Pa
1 cal	= 4.184 J
1 L·atm	= 101.325 J
1 eV	= 1.602×10^{-19} J
1 lb	= 453.6 g
1 ton	= 2000 lbs
1 tonne	= 1000 kg
1 in	= 2.54 cm

Water Data

T_{mp}	= 0°C
T_{bp}	= 100°C
C_{ice}	= 2.09 J/gK
C_{water}	= 4.184 J/gK
C_{steam}	= 2.03 J/gK
ΔH_{fus}	= 334 J/g
ΔH_{vap}	= 2260 J/g
K_f	= 1.86°C/m
K_b	= 0.512°C/m

Selected Equilibrium Constants

K_w H_2O	1.0×10^{-14}
K_a HF	7.1×10^{-4}
K_{sp} TlBr	3.4×10^{-6}
K_{sp} AgCl	1.8×10^{-10}
K_{sp} FeS	6.0×10^{-19}
K_{sp} CuI	1.1×10^{-12}
K_{sp} ZnS	2.0×10^{-25}

Selected Bond Energies

Xe-F = 131 kJ/mol
F-F = 159 kJ/mol

Selected Additional Data

$D_{methanol}$ = 0.791 g/mL
$C_{p\ methanol}$ = 81.6 J/molK
$bp_{methanol}$ = 65.0 C
$\Delta H_{v\ methanol}$ = 39.2 kJ/mol